ECE-241
Computer Systems Engineering
Project Report
23-5-2026
IoT-Based Health Monitoring System

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ABSTRACT

This project presents the design and implementation of a real-time IoT-Based Health Monitoring System capable of measuring and transmitting vital signs including heart rate, blood oxygen saturation (SpO_2), body temperature, ECG signals, and fall detection. The system is built around the ESP WROOM32 microcontroller, which integrates multiple sensors — the MAX30102 for heart rate and SpO_2, the DS18B20 for temperature, the AD8232 for ECG acquisition, and the MPU6050 for motion and fall detection. The collected data is transmitted over Wi-Fi and displayed on a web-based clinical dashboard accessible via a browser through the device's IP address, enabling real-time remote monitoring. The system also includes a buzzer-based alert mechanism that triggers upon detecting abnormal readings, ensuring timely response to critical health conditions. Results demonstrate accurate sensor readings and reliable wireless communication, confirming the system's viability as a low-cost, real-time patient monitoring solution.

Keywords: IoT, Health Monitoring, ESP32, Heart Rate, SpO_2, ECG, Temperature, Fall Detection, Wi-Fi Dashboard, Embedded Systems

SECTION I: INTRODUCTION

The rapid growth of Internet of Things (IoT) technologies has opened transformative opportunities in the healthcare sector. Traditional patient monitoring systems are largely confined to clinical environments and require expensive, bulky equipment. IoT-based health monitoring systems offer a cost-effective, portable alternative that enables continuous, real-time tracking of vital signs from virtually any location.

This project aims to design and implement a complete IoT health monitoring system that collects processes and displays patient vital signs in real time. The system integrates hardware sensors with an embedded microcontroller and a web-based user interface, forming a full-stack computer systems solution. It demonstrates the convergence of hardware design, embedded programming, networking, and data visualization — core competencies of the Computer Systems Engineering discipline.

The monitored parameters include heart rate and blood oxygen level (SpO_2) via a MAX30102 optical sensor, body temperature via a DS18B20 digital sensor, electrocardiogram (ECG) signal, and motion and fall detection via an MPU6050 accelerometer/gyroscope. An OLED display provides local feedback, while a Wi-Fi-enabled web dashboard provides remote visualization and alert notifications.

The system addresses key challenges including multi-sensor data fusion, real-time wireless transmission, and reliable alert generation. By combining all these elements, the project demonstrates how a unified computer system can be built to solve real-world healthcare monitoring needs.

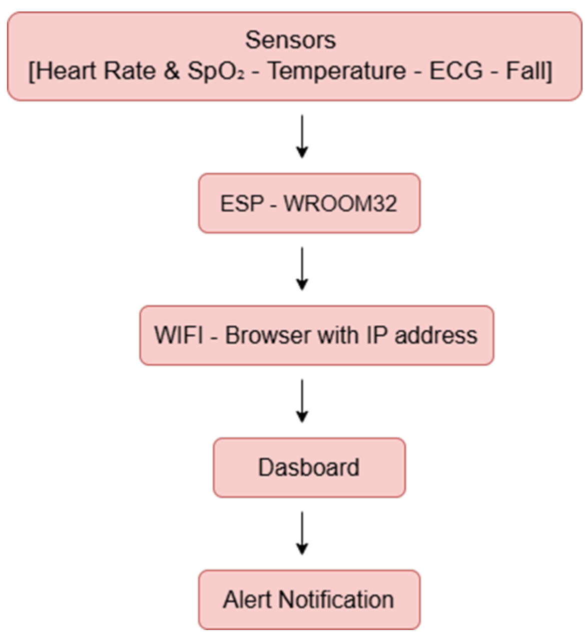

Fig. 1: System Architecture Diagram

SECTION II: BACKGROUND INFORMATION

2.1 IoT in Healthcare

2.1.1 Introduction

IoT in healthcare refers to the network of medical devices and applications connected via the internet to collect and exchange patient data. These systems enable remote patient monitoring, early detection of health deterioration, and improved clinical decision-making. Wearable and embedded medical devices equipped with biosensors can continuously track physiological parameters, reducing the need for in-person hospital visits.

The use of microcontrollers such as the ESP32 which integrates dual-core processing, Wi-Fi, and Bluetooth capabilities on a single chip has significantly lowered the cost and complexity of IoT health systems. The ESP32 serves as the central processing unit of this system, orchestrating data acquisition from all sensors and managing wireless communication.

2.1.2 Key Sensing Technologies

Each sensor in this system operates on distinct physical principles. The MAX30102 uses photoplethysmography (PPG), emitting red and infrared light through the skin and measuring reflected intensities to compute heart rate and SpO_2. The DS18B20 uses a single-wire digital protocol to report temperature with 0.5°C accuracy. The AD8232 is a dedicated ECG front-end IC that amplifies and filters biopotential signals measured through surface electrodes. The MPU6050 contains a MEMS-based accelerometer and gyroscope, enabling accurate detection of orientation changes and sudden accelerations indicative of falls.

2.2 Communication and Dashboard

The ESP32 hosts a lightweight HTTP web server that serves an HTML/JavaScript dashboard to any browser on the same local network. The browser accesses the dashboard via the device's IP address, making the system infrastructure-free beyond a Wi-Fi router. Sensor data is transmitted as JSON payloads and visualized using real-time charts. Alert notifications are generated when readings exceed configurable thresholds, and audible alerts are activated through buzzer connected to the ESP32.

SECTION III: METHODOLOGY

3.1 System Design and Architecture

The system follows a layered architecture as illustrated in Fig. 1. At the sensing layer, four distinct sensor modules collect patient physiological data continuously. The acquired signals are fed into the ESP32 (WROOM-32 module), which performs analog-to-digital conversion, signal conditioning, and preliminary data processing. The processed data is then transmitted over Wi-Fi to a browser-based dashboard, where it is visualized in real time. Alert notifications are generated automatically when readings cross predefined clinical thresholds.

3.2 Hardware Components

The hardware system was assembled on a breadboard. Table 1 below summarizes the main components and their respective functions:

Components	Function
ESP32	The main microcontroller that reads sensor data, processes it, and sends it through Wi-Fi to the cloud dashboard.
MAX30102 Sensor	Measures heart rate (BPM) and blood oxygen saturation (SpO_2).
DS18B20 Temperature Sensor	Measures body temperature in real time.
AD8232 ECG Module	Measures and displays the electrical activity of the heart (ECG signal).
MPU6050 Sensor	Detects motion, orientation, and possible falls using accelerometer and gyroscope data.
OLED Display	Displays live sensor readings locally on the device.
Buzzer	Generates alarm sounds during abnormal conditions such as low SpO_2 or high temperature.
Li-Ion Battery	Provides portable power supply for the system.

Table 1: Hardware Components and Functions

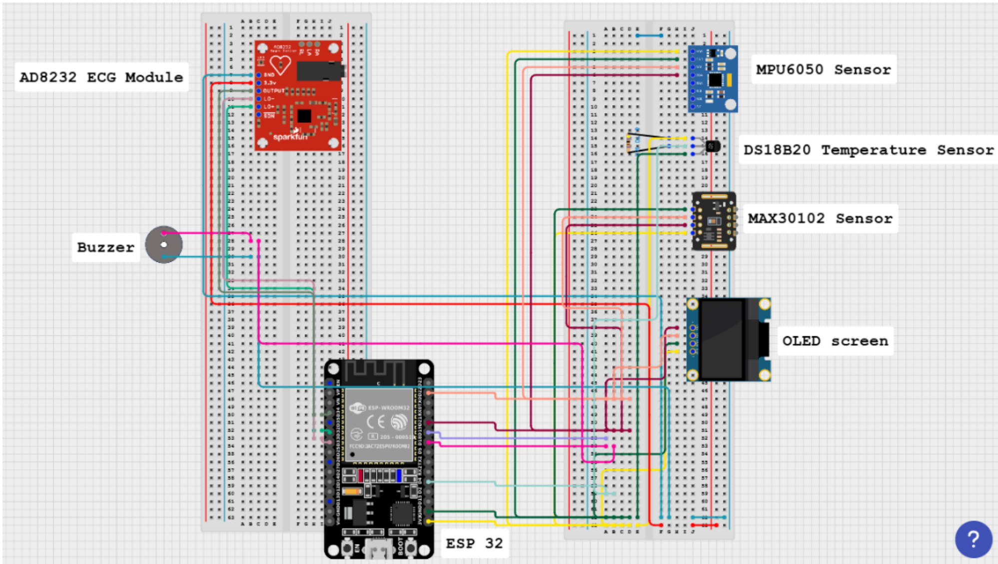

Fig. 2: Circuit Schematic – Full Hardware Assembly in Cirkit Designer

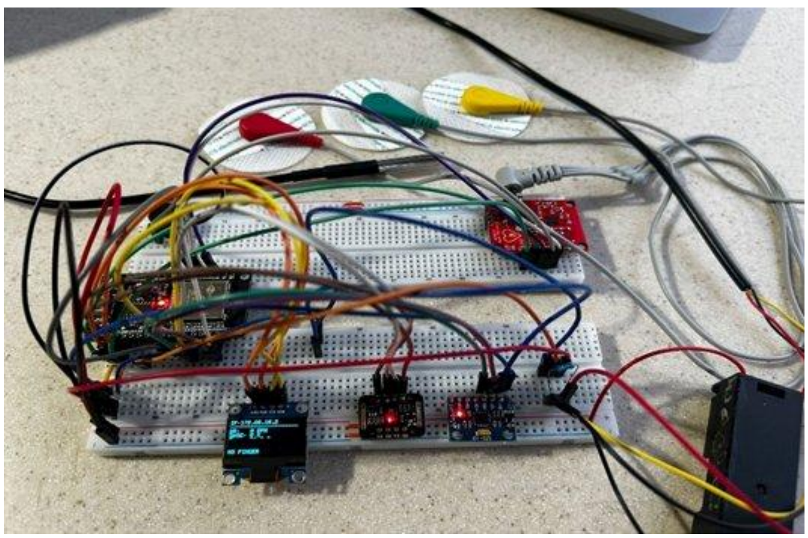

Fig. 3: Physical Hardware Prototype Assembled on Breadboard

3.3 Sensor Integration and Signal Acquisition

3.3.1 Heart Rate and SpO_2 (MAX30102)

The MAX30102 sensor uses photoplethysmography to detect volumetric changes in blood flow. Red (660 nm) and infrared (880 nm) LEDs illuminate the fingertip, and a photodetector captures the reflected intensities. The SpO_2 is calculated from the ratio of pulsatile to non-pulsatile components at both wavelengths. Heart rate is extracted by detecting peaks in the IR waveform. The sensor communicates with the ESP32 via the I^2C bus at address 0x57.

3.3.2 Body Temperature (DS18B20)

The DS18B20 digital thermometer communicates over a single-wire interface and provides 12-bit temperature resolution (0.0625°C). The sensor was placed in contact with the patient's skin. The ESP32 uses the OneWire and DallasTemperature libraries to poll the sensor and retrieve readings in degrees Celsius.

3.3.3 ECG Signal (AD8232)

The AD8232 ECG module amplifies the small biopotential signals (millivolt range) picked up by three skin-adhesive electrodes placed on the wrist and chest. The module's OUTPUT pin is connected to an analog input (GPIO pin) of the ESP32, which samples the signal via its 12-bit ADC at a rate sufficient to capture the characteristic PQRST waveform of the cardiac cycle. Lead-off detection pins (LO+ and LO-) are monitored to ensure electrode contact before recording.

3.3.4 Fall Detection (MPU6050)

The MPU6050 provides 3-axis acceleration and 3-axis gyroscope data via I²C. A fall detection algorithm computes the resultant acceleration magnitude. A sudden spike exceeding a threshold (typically 2–3g for a free-fall followed by impact) triggers a fall alert. Normal motion is classified based on the magnitude remaining within a baseline range.

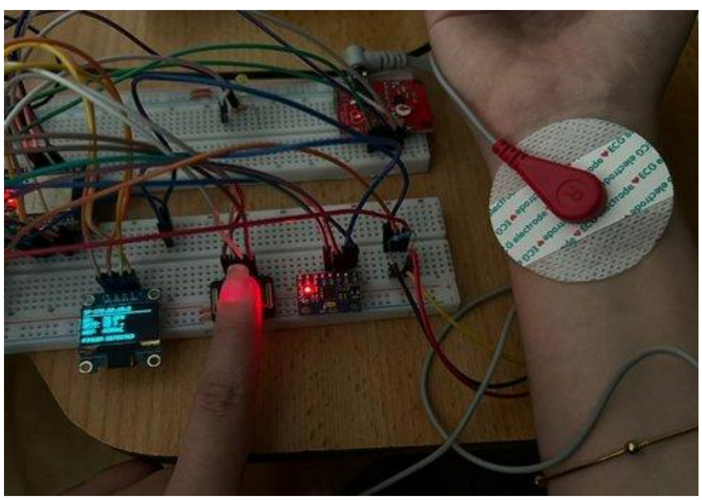

Fig. 4: Demonstration of Sensor Contact – ECG Electrode Attached to Wrist; MAX30102 Finger Placement

3.4 Firmware and Software

The software architecture of the IoT-Based Health Monitoring System is designed for real-time monitoring and data communication. The ESP32 microcontroller collects data from the heart rate, SpO_2, temperature, fall and motion sensor, and ECG sensors, processes the readings, and checks for abnormal values. The processed data is then transmitted through Wi-Fi using an IP address that connects to a dashboard for remote monitoring. An alert module is also included to generate warnings when critical health conditions are detected, such as when the temperature is higher than 38°C. The code was implemented using Arduino IDE and the libraries used were:

- **Adafruit SSD1306** and **Adafruit GFX Library** for controlling the OLED display.
- **SparkFun MAX30105** library for interfacing with the MAX30102 heart rate and SpO_2 sensor.
- **OneWire** and **DallasTemperature** libraries for reading data from the DS18B20 temperature sensor.
- **MPU6050** library by Electronic Cats for accessing accelerometer and gyroscope data from the fall detection module.

The web dashboard was built with HTML5, CSS3, and vanilla JavaScript. It fetches JSON data from the ESP32's /data endpoint every second using the Fetch API and updates both numeric displays and real-time charts rendered with the Chart.js library. The ECG chart plots the raw ADC values to visualize the cardiac waveform.

SECTION IV: RESULTS & DISCUSSION

The system was tested on a healthy adult subject and performed reliably across all measurement modalities. The web dashboard was accessible from a laptop on the same Wi-Fi network by navigating to the ESP32's assigned IP address (172.20.10.2 during testing).

4.1 Vital Signs Dashboard

Fig. 5 shows the main clinical monitor dashboard, which displays heart rate, oxygen saturation (SpO_2), temperature, fall status, and motion classification in real time. During the test session, the recorded values were: heart rate of 96 BPM, SpO_2 of 90%, and body temperature of 28.4°C. The fall indicator displayed "SAFE" and motion was classified as "NORMAL".

Fig. 5: Clinical Monitor Dashboard – Real-Time Vital Signs and Live Heart Rate Chart

The live heart rate chart updates continuously, showing the BPM trend over the most recent time window. The waveform shows physiologically plausible oscillations consistent with a resting heart rate.

4.2 ECG Signal Visualization

Fig. 6 shows the scrolling dashboard view including both the live heart rate BPM chart and the live ECG signal chart. The ECG waveform captured by the AD8232 module shows discernible cardiac cycles with QRS complex features visible, confirming proper electrode contact and adequate signal amplification.

Fig. 6: Live ECG Signal and Heart Rate Charts on the Clinical Dashboard

The ECG chart plots raw ADC values (0–4095 range from the 12-bit ADC) and updates in real time. The characteristic peaks corresponding to R-waves are identifiable, demonstrating the system's ability to capture and transmit biopotential signals wirelessly.

4.3 Alert and Notification System

The system successfully triggered alert notifications when sensor values exceeded predefined thresholds. The onboard LED flashed and the buzzer sounded when simulated abnormal conditions were introduced. The dashboard reflected the alert state immediately. This demonstrates the system's capability for automated clinical decision support.

4.4 Discussion

The results confirm that the ESP32 is capable of simultaneously managing multiple sensor interfaces, running a web server, and serving real-time data to a browser-based client — all within its dual-core architecture. The use of the I^2C bus for MAX30102 and MPU6050 reduces wiring complexity while the dedicated analog input for the AD8232 preserves ECG signal fidelity.

One observed limitation is the SpO_2 reading of 90%, which may reflect suboptimal finger placement or motion artifact during the test. Similarly, the temperature reading of 28.4°C likely reflects surface skin temperature rather than core body temperature, which is a known limitation of non-invasive surface measurement.

The system latency from sensor acquisition to dashboard update was approximately 1 second, determined by the HTTP polling interval. This is adequate for non-emergency monitoring but could be reduced to using WebSocket communication for applications requiring lower latency.

SECTION V: CONCLUSION

This project successfully demonstrates the design and implementation of a real-time IoT-based health monitoring system integrating four distinct sensing modalities — heart rate and SpO_2, body temperature, ECG, and fall detection — on the ESP32 microcontroller platform. The system satisfies all project requirements: real-time data acquisition from at least four sensors, embedded microcontroller processing, wireless communication over Wi-Fi, a web-based monitoring dashboard, and automated alert generation.

The hardware prototype was assembled on a breadboard and tested with actual sensor-to-skin contact, yielding physiologically consistent readings. The web dashboard provided accessible, real-time visualization of all vital signs from a standard browser without requiring any additional software installation. The ECG waveform visualization confirmed the system's capability to capture and transmit biopotential signals wirelessly.

Future improvements could include: upgrading to WebSocket-based communication to reduce dashboard latency below 100 ms; implementing digital filtering algorithms on the ESP32 to improve ECG signal quality and reduce motion artifacts in the MAX30102; integrating cloud-based data logging (e.g., MQTT to a cloud broker such as AWS IoT or EMQX) for long-term trend analysis; adding a rechargeable battery module for portability; and incorporating machine learning-based arrhythmia detection directly on the device using TensorFlow Lite for Microcontrollers.

The project demonstrates the viability of low-cost embedded IoT systems as accessible health monitoring solutions, with significant potential applications in remote patient monitoring, elderly care, and resource-limited clinical environments.

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Appendix

- The schematic, implementation code, and demonstration video for the IoT-based health monitoring system can be accessed via the following link:

  https://drive.google.com/drive/folders/1J-zskdFN1xD7Kmg5a4j3Q3N2h0VAG89r?usp=drive_link